StyleNova – Fashion Products E-Commerce Order Management System

Objectives

• Apply COUNT(), SUM(), AVG(), MIN(), and MAX().

• Generate total sales reports for StyleNova orders.

• Find top customers based on purchase amount.

• Identify best-selling fashion products.

• Perform category-wise sales analysis.

Aggregate functions are SQL functions used to perform calculations on a group of rows and return a single result. In StyleNova, these functions can summarize order and sales information.

1.Aggregate functions

COUNT()

```
SQL> SELECT COUNT(Order_ID) AS Total_Orders FROM Orders;

TOTAL_ORDERS
------------
           3
```

SUM()

```
SQL> SELECT SUM(QUANTITY) AS TOTAL_QUANTITY
  2  FROM Orders;

TOTAL_QUANTITY
--------------
             4
```

AVG()

```
SQL> SELECT AVG(ORDER_AMOUNT) AS AVERAGE_ORDER_AMOUNT
  2  FROM (
  3      SELECT O.ORDER_ID,
  4             SUM(O.QUANTITY * P.PRICE) AS ORDER_AMOUNT
  5      FROM ORDERS O
  6      JOIN PRODUCT P
  7      ON O.PRODUCT_ID = P.PRODUCT_ID
  8      GROUP BY O.ORDER_ID
  9  );

AVERAGE_ORDER_AMOUNT
--------------------
          1315.33333
```

MIN()

```
SQL> SELECT MIN(ORDER_AMOUNT) AS MINIMUM_ORDER_AMOUNT
  2  FROM (
  3      SELECT O.ORDER_ID,
  4             SUM(O.QUANTITY * P.PRICE) AS ORDER_AMOUNT
  5      FROM ORDERS O
  6      JOIN PRODUCT P
  7      ON O.PRODUCT_ID = P.PRODUCT_ID
  8      GROUP BY O.ORDER_ID
  9  );

MINIMUM_ORDER_AMOUNT
--------------------
                 599
```

MAX()

```
SQL> SELECT MAX(ORDER_AMOUNT) AS MAXIMUM_ORDER_AMOUNT
  2  FROM (
  3      SELECT O.ORDER_ID,
  4             SUM(O.QUANTITY * P.PRICE) AS ORDER_AMOUNT
  5      FROM ORDERS O
  6      JOIN PRODUCT P
  7      ON O.PRODUCT_ID = P.PRODUCT_ID
  8      GROUP BY O.ORDER_ID
  9  );

MAXIMUM_ORDER_AMOUNT
--------------------
                2598
```

2.Generate Total Sales Reports

A total sales report shows the revenue generated by all completed or recorded StyleNova orders. The SUM() function is applied to the Total_Amount column.

```
SQL> SELECT SUM(O.QUANTITY * P.PRICE) AS TOTAL_SALES
  2  FROM ORDERS O
  3  JOIN PRODUCT P
  4  ON O.PRODUCT_ID = P.PRODUCT_ID;

TOTAL_SALES
-----------
       3946
```

Customer-wise Sales Report

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.CUSTOMER_NAME,
  3         SUM(O.QUANTITY * P.PRICE) AS TOTAL_PURCHASE
  4  FROM CUSTOMERS C
  5  JOIN ORDERS O
  6  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  7  JOIN PRODUCT P
  8  ON O.PRODUCT_ID = P.PRODUCT_ID
  9  GROUP BY C.CUSTOMER_ID, C.CUSTOMER_NAME
 10  ORDER BY TOTAL_PURCHASE DESC;

CUSTOMER_ID CUSTOMER_NAME                                      TOTAL_PURCHASE
----------- -------------------------------------------------- --------------
        202 Priya                                                        2598
        203 Rahul                                                         749
        201 Aarthi                                                        599
```

Customers Based on Purchase Amount

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.CUSTOMER_NAME,
  3         SUM(O.QUANTITY * P.PRICE) AS TOTAL_PURCHASE
  4  FROM CUSTOMERS C
  5  JOIN ORDERS O
  6  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  7  JOIN PRODUCT P
  8  ON O.PRODUCT_ID = P.PRODUCT_ID
  9  GROUP BY C.CUSTOMER_ID, C.CUSTOMER_NAME
 10  ORDER BY TOTAL_PURCHASE DESC
 11  FETCH FIRST 5 ROWS ONLY;

CUSTOMER_ID CUSTOMER_NAME                                      TOTAL_PURCHASE
----------- -------------------------------------------------- --------------
        202 Priya                                                        2598
        203 Rahul                                                         749
        201 Aarthi                                                        599
```

Identify Best-Selling Products

```
SQL> SELECT P.PRODUCT_ID,
  2         P.PRODUCT_NAME,
  3         SUM(O.QUANTITY) AS TOTAL_QUANTITY_SOLD
  4  FROM PRODUCT P
  5  JOIN ORDERS O
  6  ON P.PRODUCT_ID = O.PRODUCT_ID
  7  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  8  ORDER BY TOTAL_QUANTITY_SOLD DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_QUANTITY_SOLD
-------------------
       102
Denim Jeans
                  2

       101
Cotton T-Shirt
                  1

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_QUANTITY_SOLD
-------------------

       104
Kids Hoodie
                  1
```

Product Sales Amount

```
SQL> SELECT P.PRODUCT_ID,
  2         P.PRODUCT_NAME,
  3         SUM(O.QUANTITY) AS TOTAL_QUANTITY_SOLD,
  4         SUM(O.QUANTITY * P.PRICE) AS TOTAL_SALES
  5  FROM PRODUCT P
  6  JOIN ORDERS O
  7  ON P.PRODUCT_ID = O.PRODUCT_ID
  8  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  9  ORDER BY TOTAL_SALES DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_QUANTITY_SOLD TOTAL_SALES
------------------- -----------
       102
Denim Jeans
                  2        2598

       104
Kids Hoodie
                  1         749

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_QUANTITY_SOLD TOTAL_SALES
------------------- -----------

       101
Cotton T-Shirt
                  1         599
```

3. Combined Business Sales Report

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.CUSTOMER_NAME,
  3         O.ORDER_ID,
  4         P.PRODUCT_ID,
  5         P.PRODUCT_NAME,
  6         O.QUANTITY,
  7         P.PRICE,
  8         (O.QUANTITY * P.PRICE) AS SUBTOTAL
  9  FROM CUSTOMERS C
 10  JOIN ORDERS O
 11  ON C.CUSTOMER_ID = O.CUSTOMER_ID
 12  JOIN PRODUCT P
 13  ON O.PRODUCT_ID = P.PRODUCT_ID
 14  ORDER BY O.ORDER_ID;

CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY      PRICE   SUBTOTAL
---------- ---------- ----------
       201 Aarthi                                                    301
       101
Cotton T-Shirt
         1        599        599

CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY      PRICE   SUBTOTAL
---------- ---------- ----------
       202 Priya                                                     302
       102
Denim Jeans
         2       1299       2598

CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY      PRICE   SUBTOTAL
---------- ---------- ----------
       203 Rahul                                                     303
       104
Kids Hoodie
         1        749        749
```